Structural certificates for four alternating and symmetric tables

Private L054 recovery component

Theorem 1. *Every structural description in the current companion tables for A_7, S_7, A_8, S_8 has an actual decomposition certificate in its numerical row class. These four tables contain respectively $40, 96, 137, 296$ rows, for a total of 569. A description asserts the displayed decomposition; it need not determine a unique group.*

Source and acquired premises. The tables are attributed to Thomas Connor and Dimitri Leemans. All 25 native pages were directly inspected. We acquire the previously accepted complete ambient class counts and numerical row maps, together with the accepted concrete structural-model registry. These are current companion data; historical 2013 PDF-byte identity is not asserted. The 2015 naming article specifies association from the left. We preserve this convention and every native dot, colon, cross and exponent. A dot means nonsplit extension, a colon split extension, and a cross means direct product. For example, the S_8 row 13 means $(((A_4 \times 2) \times 2) : S_3) \cdot 2$. Its outer dot is not replaced by a colon. Row 200 has 4^2, meaning $C_4 \times C_4$, and row 119 has the unsigned extraspecial atom 2^{1+4}. We assert neither plus nor minus type for that atom. The inherited Q_{12} convention is the explicit dicyclic presentation from the accepted registry, not an inferred quaternion 2-power convention.

Concrete registry and transports. The input has 88 acquired concrete models and four elementary standard models, for A_7, S_7, A_8, S_8. Different concrete models may share a structural description. The new application adds 65 models. For a standard symmetric model we use a full cycle and a transposition; for an alternating model we use $(1\,2\,i)$, $3 \le i \le n$. The literal closure has order $n!$, respectively $n!/2$, and the alternating generators are even. Thus these closures are exactly the groups of all permutations, respectively all even permutations, on the indicated point set.

Every reuse has an explicit bijection fixing identity and preserving right multiplication by every model generator. Induction on generator words proves that it is a homomorphism, hence an isomorphism. This is the acquired MC.3 criterion; no group-name equality substitutes for it. The literal checker expands candidate generator maps, then checks all 692 resulting bijections and 510,303 right-generator edges.

For a split description it checks $N \lhd H$, a named subgroup K, $N \cap K = 1$ and $|N||K| = |H|$. A colon also has a noncommuting pair of factor generators; for a cross all such pairs commute. Normality and commutation on generators imply the corresponding whole-group facts. For the new prime-index nonsplit descriptions, every outside element fails $x^p = 1$: an index-p complement exists exactly when an outside element has order p. Factor references are acyclic and already verified.

Lemma 1 (Two additional atom criteria)**.** *1. Two commuting elements of order four which generate a group of order sixteen identify that group with $C_4 \times C_4$.*

2. Suppose $|H| = 32$ and z has order two. If $Z(H) = \langle z \rangle$, every square lies in $\langle z \rangle$, and generator commutators lie in $\langle z \rangle$ with at least one equal to z, then H is extraspecial.

Proof. In the first case the defining commuting generators give a surjection from $C_4 \times C_4$ onto H, and equal orders make it an isomorphism. In the second case the normal closure of the generator commutators is H', so $H' = Z(H) = \langle z \rangle$ and $H/Z(H)$ is elementary abelian. For completeness, $\Phi(H) = H'H^2$: every maximal subgroup of a finite 2-group is normal of index two, hence contains $H'H^2$; and hyperplanes in the elementary abelian quotient $H/H'H^2$ have trivial intersection. The asserted maximal-subgroup fact follows from the normalizer condition: a proper subgroup is strictly contained in its normalizer, by induction modulo the center (if the center is not contained in the subgroup it already enlarges the normalizer). A maximal subgroup is therefore normal, and its quotient has order two. Thus $\Phi(H) = \langle z \rangle$ as well, which is precisely the extraspecial condition. □

The checker computes the center by testing every group element against every generator, checks every square, and checks all generator commutators. It does not infer an extraspecial type from order alone.

Actual embedded rows despite printed point-set names. Some retained arrays print `AlternatingGroup([points])` or `SymmetricGroup([points])`. We treat their conventional point-set generators only as candidates, without claiming that GAP's display preserved the original producer's hidden embedding. We close the ambient permutations and all candidate subgroups, check membership, and enumerate the ambient conjugacy orbit of every candidate. The resulting classes are pairwise disjoint and their number equals the acquired complete class count. They therefore cover all subgroup classes.

For efficiency each actual ambient permutation is assigned its unique closure index. Conjugation by an ambient generator is an exactly computed permutation of those indices. Subgroups become subsets of this index set; this is a bijective encoding of the same literal subgroups, preserving membership, inclusion and conjugation. No class decision is approximate.

All actual subgroups now occur in the conjugacy orbits. For each retained parent H, scan proper subgroups in descending order and retain $K < H$ exactly when it lies in no already retained larger subgroup. This gives precisely the maximal subgroups of H. If a such children belong to a class with ambient length c_K, double counting conjugate incidences gives $b = ac_H/c_K$ parents over each child. Every source row order, class length, and both weighted directions for every ordered row pair are checked. This independently pays the actual source-class assignment of every candidate, including the point-set expressions.

Finite execution and independent controls. One bounded witness pass returned all 569 candidates in 12.56 seconds. It searched only the retained groups for decomposition and transport witnesses. Its failed searches supply no proof premise. The final reader requires all source occurrences and actual maps, verifies 65 new models, 19 new direct products, one C_4^2 model and one extraspecial model of order 32. Its prime-index nonsplitting tests inspect 320 outside elements. No ambient subgroup census is repeated.

The embedded audit has the following exact finite outcomes:

Table	Classes	Actual subgroups	Edges	Row pairs
A_7	40	3786	98	1600
S_7	96	11300	290	9216
A_8	137	48337	419	18769
S_8	296	151221	1080	87616
Total	569	214644	1887	117201

The new ambient-index encoding is additionally compared against the previous implementation using literal permutations on the complete S_5, A_6 and $\mathrm{P\Sigma L}_2(9)$ fixtures. Their class counts, subgroup counts, edges, every source pair and exact operation counts agree.

Independent atom controls construct regular actions on $\mathbb{F}_2^5$ using the bilinear central cocycles

$$(x, c)(y, d) = (x + y, c + d + x_2y_1 + x_4y_3 + \epsilon(x_1y_1 + x_2y_2)), \qquad \epsilon = 0, 1.$$

Both order-32 models pass. All fifteen nonzero parity assignments on the four noncentral generators yield actual order-16 kernels, with intersection exactly the center of order two. A separate coordinate action of $\mathbb{Z}/4 \times \mathbb{Z}/4$ passes. Dependent generators, $C_4 \times C_2 \times C_2$, elementary abelian order 32, and a proposed noncentral center generator are rejected.

Damaged full certificates test map identity and generator edges, missing or duplicate occurrences, wrong embedded generators and row maps, source powers, a dot changed to a colon, association, cyclic references, complements, quotient data and both new atom witnesses. Separate embedded controls reject duplicate classes, an outside ambient subgroup, wrong class lengths, row maps and incidence weights. Unsupported named group constructors are rejected. The previously acquired prime-index, composite-quotient and direct-product small controls remain executable. Ordinary and optimized Python must give identical complete reports.

Scope. These are 569 new structural row joins to current qualified data. The acquired complete class counts and earlier concrete named models remain explicit premises. The unsigned extraspecial and inherited dicyclic conventions remain visible. No assertion of canonical extension labels, historical file identity, a conjugacy-poset lattice property, whole-source completion, question resolution, formalization or measurement follows. The present proof, implementation and applications were authored under root's instruction to reuse the concrete registry. Root reviews the finished component independently.